Ocean pout (*Zoarces americanus*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	2.84	1.83	
Complexity in Reproductive Strategy	1.96	2.68	
Dispersal of Early Life Stages	3.6	2.81	
Egg and Larval Survival and Settlement Requirements	2.68	1.89	
Historical Variability in Temperature	2.76	2.9	
Juvenile Habitat Requirements	1.92	2.9	
Other Stressors	2.6	2.4	
Population Growth Rate	2.84	2.54	
Prey Specificity	1.96	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.72	2.5	
Spawning Cycle	2.6	2.66	
Stock Size/Status	4	2.32	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Bottom Temperature	2.49	1.07	
Bottom Oxygen	2.89	0.96	
Bottom Salinity	2.62	1.00	
Bottom Aragonite Solubility	1.16	0.53	
Net primary production	1.03	0.11	
Downward Flux of Particulate Organic Carbon	1.38	0.47	
Total Exposure	**Moderate (2.09)**	**0.91**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Moderate (6.26)**	**2.72**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Ocean pout. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Ocean pout is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

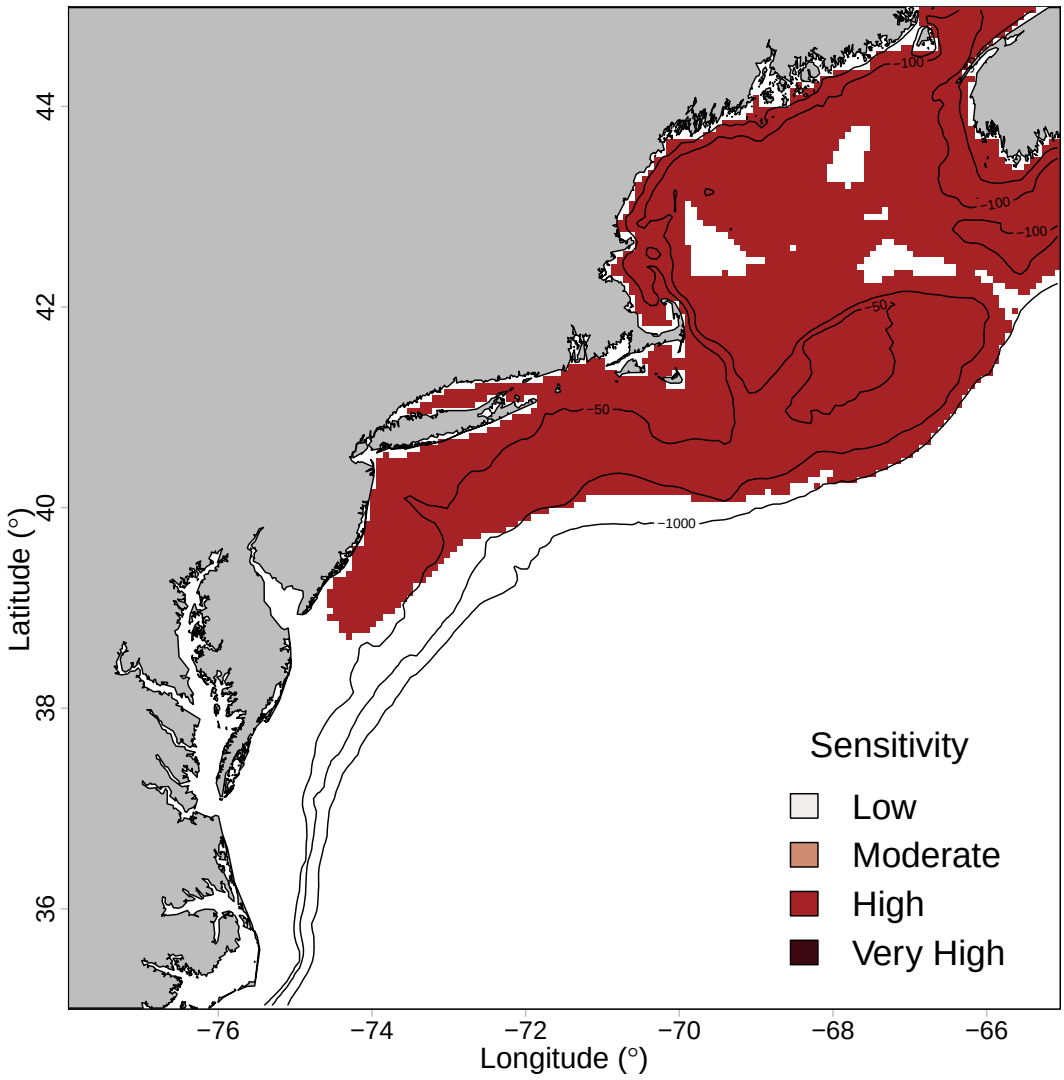


Figure 2: Ocean pout Sensitivity on the Northeast Shelf

The ocean pout received a **High** sensitivity score, with a certainty of 100%. Two attributes had scores greater than 3: Stock Size (4) and Dispersal of Early Life Stages (3.6) (Figure 1). The stock is currently overfished, but overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2022). Eggs masses and larvae are both demersal, with larvae quickly associating with benthic habitats after hatching (Steimle et al. 1999).

In the 2016 Climate Vulnerability Assessment, ocean pout also received a High sensitivity score. This was driven by Dispersal and Early Life History, Stock Status, and Population Growth Rates (Hare et al. 2016). In our assessment, Population Growth Rate received a score of 2.8.

Exposure:

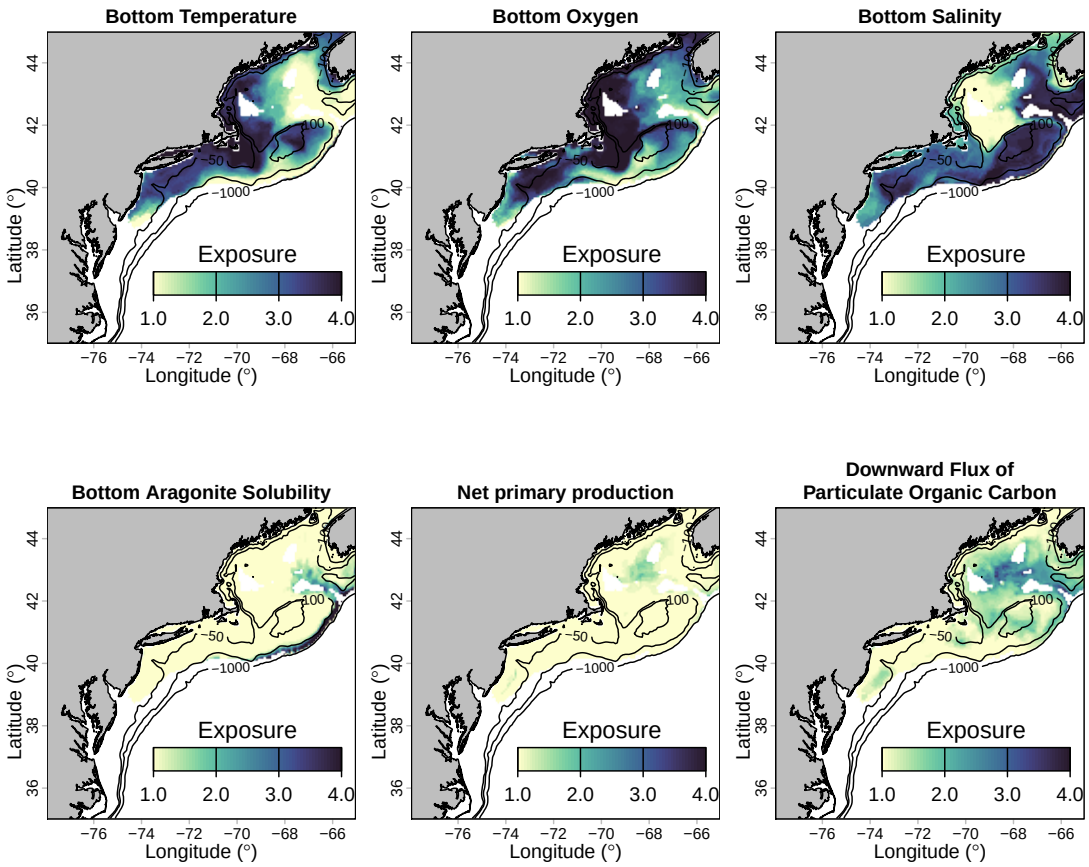


Figure 3: Maps of each Exposure Factor for Ocean pout. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure scores for ocean pout were **Moderate** (2.09 +/- 0.91) (Figure 1). Scores were High in Southern New England, along the coast of the Gulf of Maine, and on Georges Bank, and Low throughout the rest of the Gulf of Maine (Figure 4). These patterns were driven by bottom temperature, oxygen, and salinity exposures (Figure 3).

In the previous Assessment, ocean pout received a High exposure score, driven by high sea surface temperature and ocean acidification scores (Hare et al. 2016).

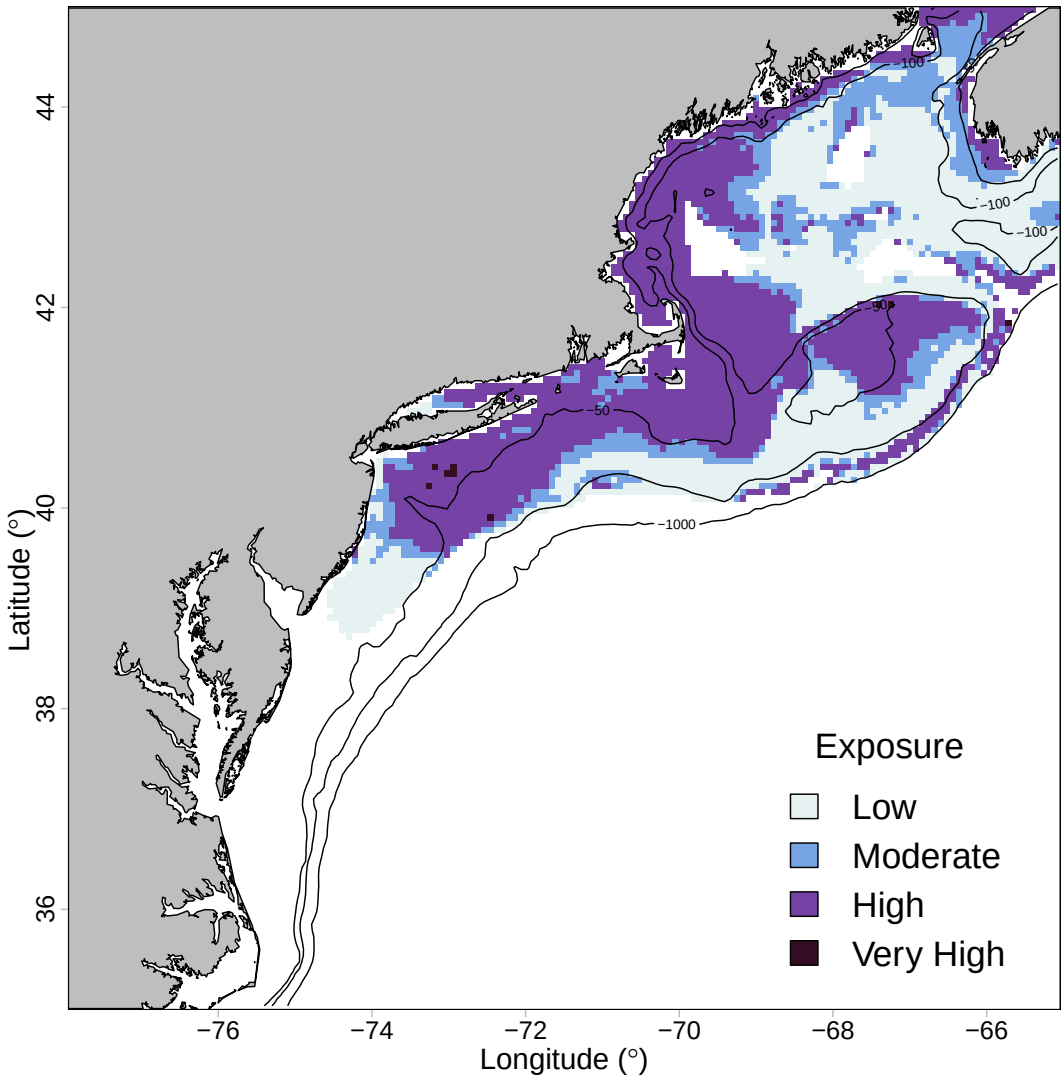


Figure 4: Total Exposure on the Northeast Shelf for Ocean pout. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

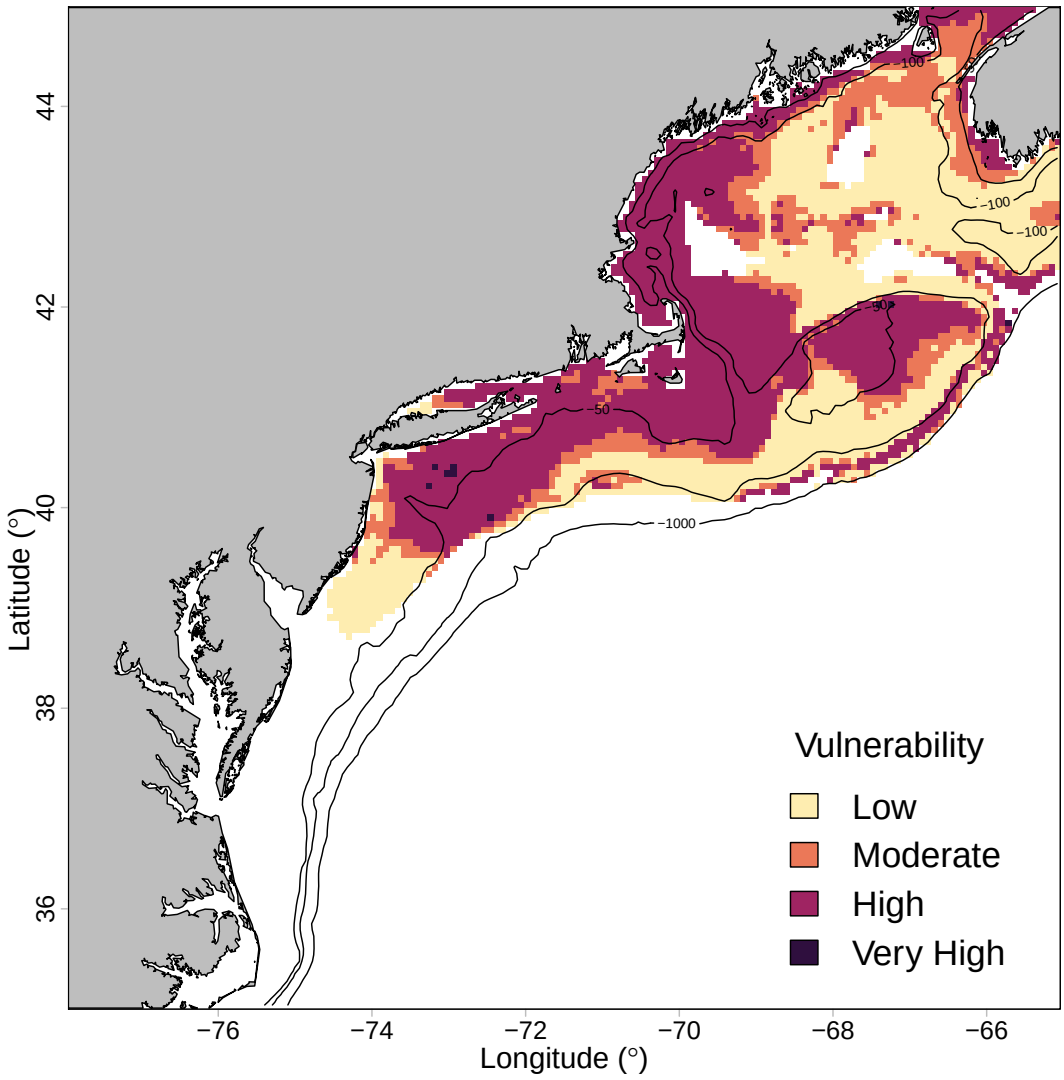


Figure 5: Vulnerability on the Northeast Shelf for Ocean pout. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Ocean pout had a **Moderate** mean Vulnerability score (6.26 +/- 2.72) (Figure 1). Vulnerabilities were High in Southern New England, Georges Bank, and along most of the coast of the Gulf of Maine, but Low elsewhere (Figure 5).

In the previous Assessment, ocean pout received a High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

83% of data quality scores were greater than or equal to 2, indicating strong data quality. Adult Mobility and Egg and Larval Survival and Settlement Requirements were considered data gaps (mean attribute

data quality score < 2) (Figure 1).

Model Confidence Score:

The ocean pout ensemble model received a mean model confidence score of 2 (+/- 0.4), indicating moderate confidence in the model. The predicted distributions aligned with experts- expectations. The model tended to underpredict present on Georges Bank, particularly in spring, which drove down scores. The ensemble model AUC score was 0.87, indicating good model performance.

Additional Metrics

Distribution Shifts:

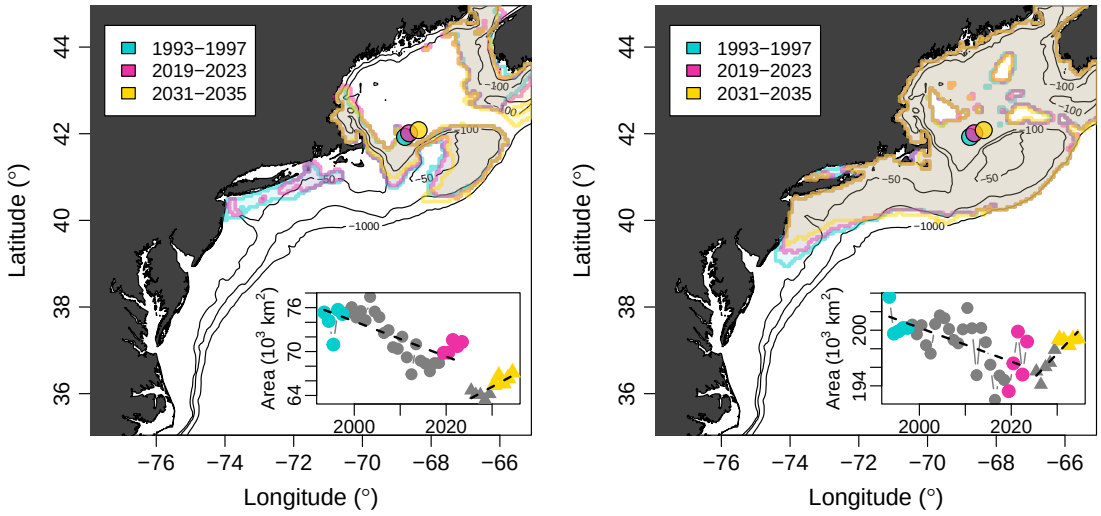


Figure 6: Distribution shifts on the Northeast Shelf for Ocean pout. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Ocean pout experienced significant declines in habitat area between 1993 and 2023. This was driven by loss of core habitat south of Long Island and the northward movement of the southern edge of their range (Figure 6). During this time period, the weighted centroid shifted 14 km northeast.

Under near future climate conditions, the range of ocean pout is predicted to expand. Core habitat is also expected to expand, although it will still be smaller than 2023 levels (Figure 6). The weighted centroid is predicted to shift a further 24 km east-northeast.

In the previous Assessment, ocean pout received a Moderate Distributional Vulnerability Rank due to their preference for complex habitats (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	74,722	201,136	4.43	7.62
2019-2023	13.6 km NE	70,825 (-3,897)	196,853	4.24 (-0.19)	7.37 (-0.24)
2031-2035	24.3 km ENE	66,247 (-4,579)	198,933 (+2,080)	4.24 (0)	7.30 (-0.08)

Directional Effect:

Experts predict that ocean pout will be **negatively** impacted by climate change with 99% certainty. Ocean pout prefer colder waters and are sluggish movers (Steimle et al. 1999), which may make it difficult to avoid stressful conditions such as hypoxia (Azarovitz et al. 1976). In the previous Climate Vulnerability Assessment, experts made a similar prediction, but certainty was lower (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Nye et al. (2009) identified an increase in depth of occurrence of ocean pout in the Gulf of Maine. Although ocean pout have been reported to be quite hardy, they appear to have limited tolerance of hypoxic and anoxic conditions; they are common victims of these conditions which occasionally occur in the New York Bight (Azarovitz et al. 1976). Ocean pout are sluggish swimmers, which can explain their inability to avoid stressful conditions, such as hypoxia (Steimle et al. 1999). Killen et al. (2008) found that young ocean pout from Newfoundland do not show thermal compensation in response to long-term temperature changes.

Life History Synopsis

Ocean pout are a non-migratory, bottom dwelling fish found along the Atlantic coast from Labrador to Delaware (Steimle et al. 1999). Spawning occurs in the late summer through early winter (peak in September-October) with earlier peaks (August-October) in the south (Wilk and Morse 1979). Eggs are laid in gelatinous masses in a sheltered place on the bottom, such as rocky crevices, where they are guarded either by one or both parents until hatching (Steimle et al. 1999). Eggs develop in 2-3 months, temperature dependent (Olsen and Merriman 1946). Larvae are about 30 mm long at hatching (Clark and Livingstone 1982) and live at the bottom at temperatures below 10°C (Steimle et al. 1999). Juvenile and adult ocean pout use both open and rough habitats. Sedberry (1983) reported that juveniles were common during the summer on the sandy, mid- to outer-continental shelf (approximately 35-95 m). In the Gulf of Maine, juveniles occur in shallow coastal waters around rocks and attached algae, and in rivers with saline bottom waters (Auster 1985). MacDonald (1983) reported that ocean pout feed by sorting mouthfuls of sediments for infauna and do not use visual cues to stimulate feeding; they maintained that ocean pout were not very effective at capturing moving prey and rarely left the bottom to feed. Auster (1985) and Auster et al. (1995), however, considered ocean pout an "ambush predator" that waits in sediment depressions for prey to approach or drift by in the current. Juveniles feed primarily on gammarid amphipods and polychaetes (Sedberry 1983). Adult ocean pout feed on a variety of benthic invertebrates, including polychaetes, mollusks, crustaceans, and echinoderms (Steimle et al. 1999). Adults reach maturity

at 2+ years (Steimle et al. 1999), with a maximum age of 18 years (Hoenig 1983). Although they do not migrate, ocean pout move seasonally within a region: offshore in late summer and inshore in the spring in the Gulf of Maine, and offshore to cooler waters in the summer returning inshore in the fall in Southern New England and Georges Bank (Steimle et al. 1999). Ocean pout are managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. As of the 2022 management track assessment, the stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2022).

Supplemental Scores

Exposure Timeseries

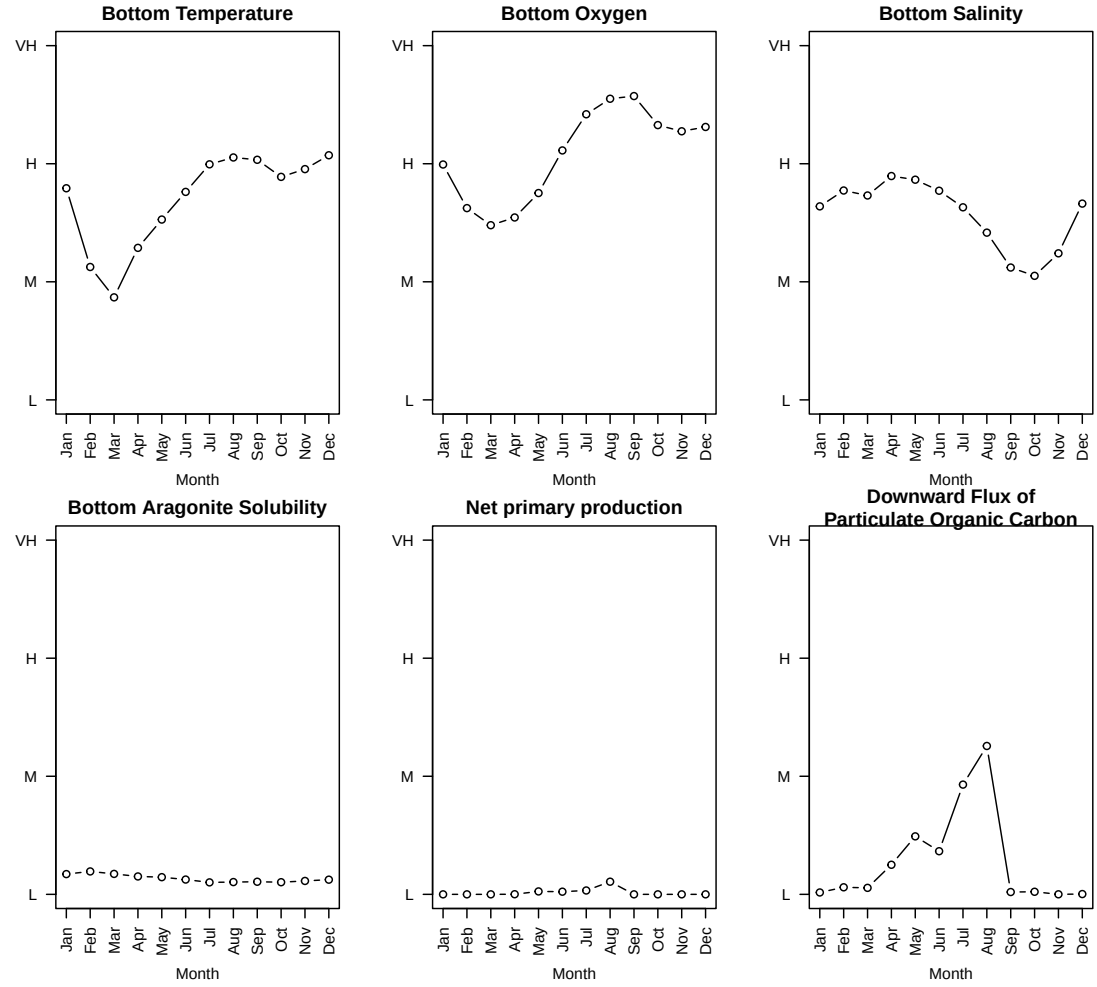


Figure 7: Monthly Timeseries of each Exposure Factor for Ocean pout. Exposure was calculated between 2014-2023 and 2025-2035

Monthly Exposure ranged from Low to High. Scores were Low in March and High in August and September (Figure 8). These patterns were driven by bottom temperature, bottom oxygen, and downward particulate carbon flux (Figure 7).

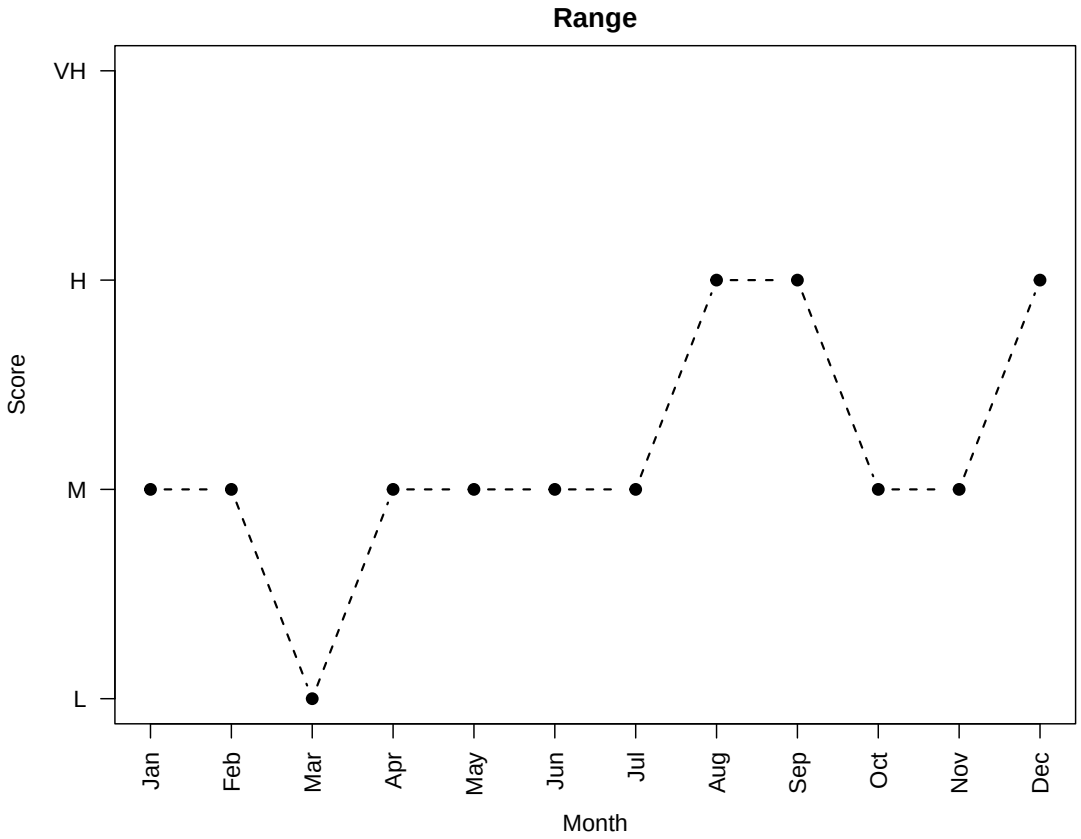


Figure 8: Timeseries of Total Exposure for Ocean pout

Exposure due to Significant Factors

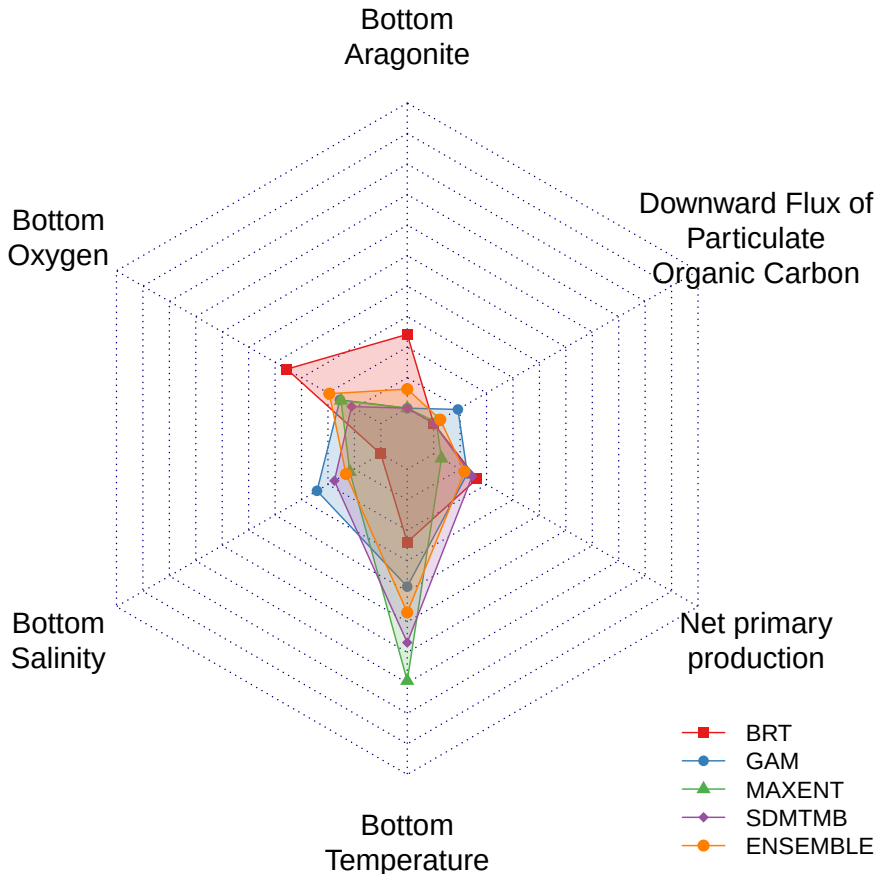


Figure 9: Dynamic Factors driving Ocean pout distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, salinity, and net primary productivity were important dynamic drivers of ocean pout distributions (Figure 9). Ocean pout are demersal fish that filter feed on bottom sediments (Steimle et al. 1999). When only these factors were used to calculate exposure, mean scores remained Moderate (2 +/- 0.92) (Figure 11, Figure 10).

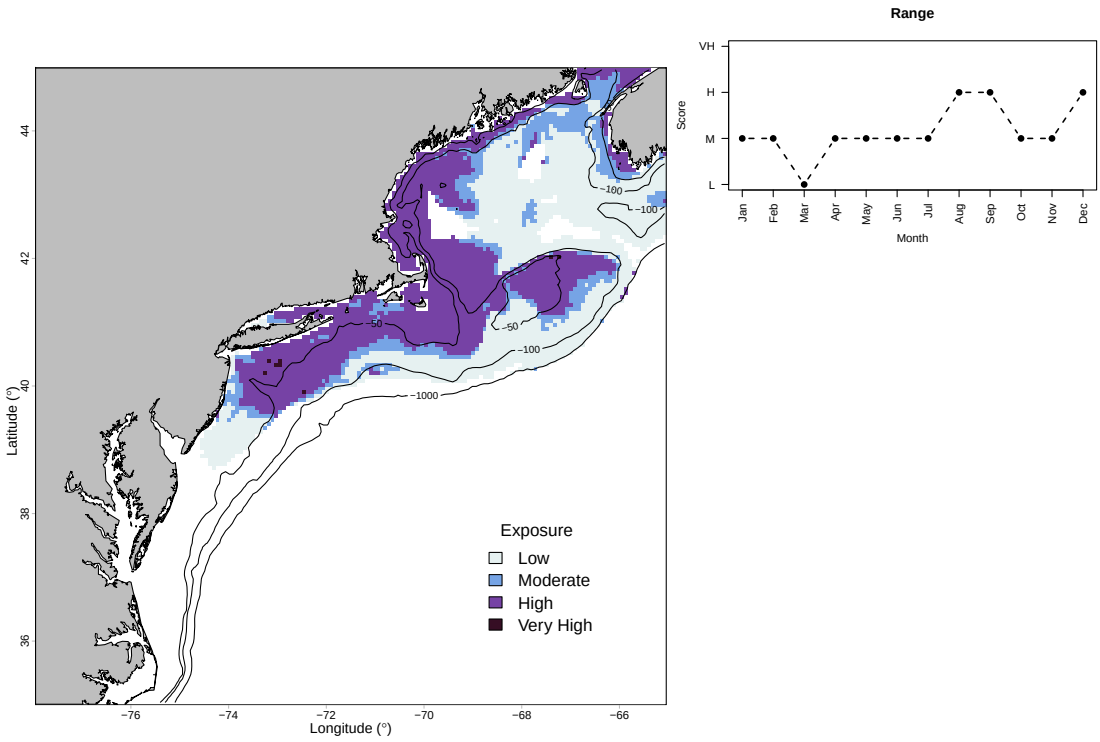


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Ocean pout distributions

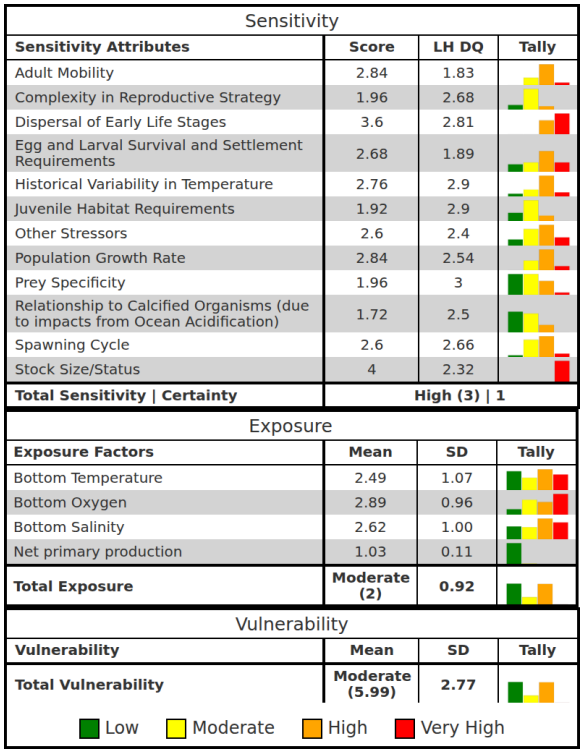


Figure 11: As in Figure 1 for Ocean pout, but with only significant Exposure factors and the resulting Vulnerability Scores.

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